From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST

FIITJEE

JEE(Advanced)-2014

ANSWERS, HINTS & SOLUTIONS

PART TEST – IV

(Paper – 2)

ANSWERS KEY

	PHYSICS	CHEMISTRY	MATHEMATICS
Q. No.	ANSWER	ANSWER	ANSWER
1.	C	D	A
2.	B	D	D
3.	D	C	A
4.	C	A	C
5.	D	D	D
6.	C	A	A
7.	A	B	C
8.	D	A	B
9.	A	B	B
10.	A	A	C
11.	D	A	B
12.	A	A	B
13.	C	B	D
14.	A	B	D
15.	B, C, D	A, B	A, B, C, D
16.	A, B, C, D	A, C	A, B
17.	A, B, C, D	A, B, D	A, B, C
18.	A, B, C	B, C	B, C, D
19.	A, C	D	A, C, D
20.	A, B	A, B	A, B, C, D

Physics

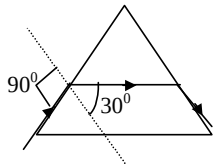
PART – I

2. $v_{Ix} = -\frac{v^2}{u^2}v_{0x}$

$v_{IY} = -\frac{v}{u}v_{OY}$, $v_{iZ} = -\frac{v}{u}v_{OZ}$

3. $\frac{\omega}{f}+\frac{\omega'}{f'}=0$

4. $\mu = \frac{\sin 90^0}{\sin 30^0} = 2$



5. $I_{max} - I_{min} = \frac{1}{3}I_{max} \Rightarrow I_{min} = \frac{2}{3}I_{max}$

Further, $\frac{I_{max}}{I_{min}} = \left(\frac{A_1+A_2}{A_1-A_2}\right)^2$

6. $A = C\sqrt{r}$

from Bohr's theory mvr = constant

and $v \propto \frac{1}{\sqrt{r}}$

$\therefore$ $L \propto vr$

$\therefore$ A is angular momentum

7. $Q = \frac{\omega_0 L}{R} = \frac{\omega_0}{2\Delta\omega}$

$\therefore$ $2\Delta\omega = R/L = f_1 - f_2$

8. $\frac{I_{min}}{I_{max}} = \frac{(A_1-A_2)^2}{(A_1+A_2)^2} = \left[\frac{0.2-0.16}{0.2+0.16}\right]^2 = \left(\frac{1}{9}\right)^2 = \frac{1}{81}$

9-10. Consider the force on electron $e\omega rB - eE = m\omega^2 r$

$\therefore$ $E = \left(\omega B - \frac{m\omega^2}{e}\right)r$

$\therefore$ $\Delta V = \left(\omega B - \frac{m\omega^2}{e}\right)\frac{R^2}{2}$

Considering a Gaussian surface of cylindrical shape having length ℓ, inner radius r and outer radius (r + dr)

$(E + dE)2\pi r\ell - (E)2\pi r\ell = \frac{\rho 2\pi r\ell dr}{\varepsilon_0}$

$\frac{\rho}{\varepsilon_0} = \frac{dE}{dr} = \omega B - \frac{m\omega^2}{e}$

$\therefore$ $\rho = \omega\left(B - \frac{m\omega}{e}\right)\varepsilon_0$

11-12. From conservation of energy

$$\frac{q^2}{2C}+\frac{1}{2}mv^2+\frac{1}{2}kx^2=E\text{, where } q = CB\ell v$$

$$\frac{dq}{dt}=CB\ell\frac{dv}{dt}$$

After differentiating

$$\frac{d^2x}{dt^2}=-\frac{dv}{m+CB^2\ell^2}x$$

$$\therefore\quad x = x_0\cos\left(\sqrt{\frac{k}{m+CB^2\ell^2}}\right)t$$

$$v=\frac{dx}{dt}\text{ and } q = CB\ell v$$

17. $v + u = D$

and $v - u = X$

$$\Rightarrow\quad v=\frac{D+X}{2},u=\frac{D-X}{2}$$

and $$f=\frac{D^2-X^2}{4D}$$

$$m_1=\frac{D+X}{D-X},\ m_2=\frac{D-X}{D+X}$$

18. $$\frac{I_{max}}{I_{min}}=\frac{(a_1+a_2)^2}{(a_1-a_2)^2}$$

$$\frac{a_1}{a_2}=\sqrt{\frac{I_1}{I_2}}=\sqrt{\frac{4}{9}}=\frac{2}{3}$$

$$\therefore\quad \frac{I_{max}}{I_{min}}=25$$

If an identical paper is pasted across second slit shifted C.B.F. will back in central point.

$$\text{shift}=(\mu-1)\frac{tD}{d}$$

$$\text{fringe width}=\frac{n\lambda D}{d}$$

$$\therefore\quad n=\frac{\text{shift}}{\text{fringe width}}=\frac{(\mu-1)t}{\lambda}=15$$

19. Energy available for excitation = E/2

$\therefore$ if E < 20.4 eV excitation will not take place.

For E < 20.4 eV collistion is perfectly elastic and neutron must come to rest.

Chemistry PART – II

SECTION – A

1. Number of 'A' particles per unit cell $= \frac{1}{8}\times 8 + \frac{1}{2}\times 6 = 4$
Number of octahedral voids = Number of B particles = 4
Number of tetrahedral voids = Number of C particles = $2 \times 4 = 8$
Hence formula = $A_4B_4C_8 = ABC_2$
When all particles along one body diagonal are removed two A atoms from corners are removed.
Hence new 'A' particles $= \frac{1}{8}\times 6 + \frac{1}{2}6 = \frac{3}{4} + 3 = \frac{15}{4}$
Number of B particles = 3
Number of C particles = 6
Hence new formula = $A_{\frac{15}{4}}B_3C_6$
$A_5B_4C_8$

2. 98% pure copper is called blister copper.

3. $FeSO_4 + 2H_2O \longrightarrow Fe(OH)_2 + H_2SO_4$
Addition of H_2SO_4 to this solution reverses the hydrolysis of $FeSO_4$.

4. The reaction is $Au + 4HCl + 3HNO_3 \rightarrow HAuCl_4 + 3NO_2 + 3H_2O$

5. Solution Bi^{3+} will form black precipitate where as Mn^{2+} does not in acidic medium due to higher K_{sp} value.
Hg^{2+} gets precipitate as HgS but Al^{3+} does not.
Cu^{2+} gets precipitated as CuS(black) whereas Zn^{2+} and Ni^{2+} does not precipitated in acidic medium.

6. $\underset{1-\alpha}{Ca(NO_3)_2} \rightleftharpoons \underset{\alpha}{Ca^{2+}} + \underset{2\alpha}{2NO_3}$

$$\frac{P^\circ - P}{P^\circ} = \left(\frac{6.69/164}{\frac{6.69}{164} + \frac{100}{18}}\right) i$$

$$\frac{760 - 746.9}{760} = i(0.00729)$$

$$i = \frac{1+2\alpha}{1-\alpha} = 2.364$$

$\alpha = 0.68$

7. The half – cell reactions of the given cell will be
$Cr \longrightarrow Cr^{3+} + 3e \quad E^\circ = 0.74$ V
$Cu^{2+} + 2e \longrightarrow Cu \quad E^\circ = 0.34$V
$\therefore E_{Cell} = 0.74+0.34 = 1.08$V

8. $3Na_2S_2O_3 + BiCl_3 \longrightarrow Na_3[Bi(S_2O_3)_3] + 3NaCl$

$$[Bi(S_2O_3)_3]^{3-} + 3H_2O \longrightarrow Bi_2S_3 + 3H_2SO_4$$

9. $\Delta G = -nFE$
$= -2 \times 96500 \times 1.1$
$= -212300$ J

10. $(\Delta G°)_R = (G°)_P - (G°)_R$
$= -154 - 64.4 = -218.4$ kJ.
$\Delta G° = -nFE°$
$-218.4 \times 10^3 = -2 \times 96500 \times E°$
$E° = 1.13$ V.

11. Let the solution is 100 gm
$\therefore$ 25 gm is impure $MgCl_2$ ($MgCl_2$ + MCl_3)
$\therefore$ 6 gm is MCl_3

$$\Delta T_f = i \times k_f \times m$$

$$= 1.86 \times \left[\left(i \times \frac{w_B}{M_B}\right)_{MgCl_2} + \left(i \times \frac{w_B}{M_B}\right)_{MCl_3}\right] \times \frac{1000}{w_A}$$

$$= 1.86 \times \left[\left(3 \times \frac{19}{95}\right) + \left(2.5 \times \frac{6}{150}\right)\right] \times \frac{1000}{75}$$

$$= 1.86 \times (0.6 + 0.1) \times \frac{1000}{75}$$

ΔT_f = **17.36 K**

12. At –21.7°C amount of solvent can be calculated as

$$21.7 = 1.86 \times \left[\left(3 \times \frac{19}{95}\right) + \left(2.5 \times \frac{6}{150}\right)\right] \times \frac{1000}{w_A}$$

w_A = 60 gm
$\therefore$ Amount of ice formed = 75 – 60 = **15 gm**.

15. Cataphoresis is the process used for coagulation of colloids in which +ve particles move towards –ve pole and –ve particles move towards +ve pole in the presence of electric field.

16. SiO_2 has different molecular arrangements in different forms.

18. More is value of *i* lesser will be freezing point of solution and $i = \frac{\text{Molecular wt. theoretical}}{\text{Molecular wt. experimental}}$

19. The first statement is false because F_2 and Cl_2 are gases, Br_2 is liquid while I_2 is a solid. Bromine dissolves in chloroform imparting reddish-brown colour. Chlorine dioxide is evolved when $KClO_3$ is treated with conc. H_2SO_4

20. Surface tension of lyophilic aquasols is lower than that due to force of attraction between dispersed phase and dispersion medium.
Its viscosity is higher due to force of attraction between dispersed phase and dispersion medium.

Mathematics PART – III

1. Let the first chosen square is white. Then the total number of squares which do not belong to the same row/column are 49 of them 25 are white and 24 are black.

Thus corresponding probability = $\frac{1}{2}.\frac{25}{49}$

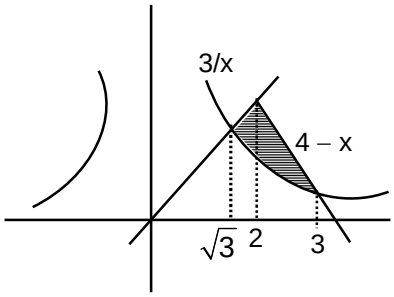
If first square is black, then corresponding probability = $\frac{1}{2}.\frac{25}{49}$

Thus required probability = $\frac{1}{2}.\frac{25}{49}+\frac{1}{2}.\frac{25}{49}=\frac{25}{49}$

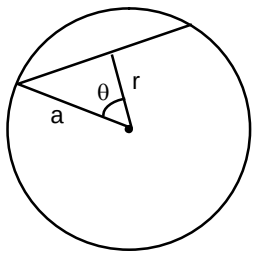
2. $A = \int_{-\pi/3}^{2\pi} |\sin x - \cos x|\,dx \Rightarrow 4\sqrt{2}+\frac{\sqrt{3}+1}{2}$.

4. The given determinant is equal to $\begin{vmatrix} |\vec{a}|^2 & \vec{a}\cdot\vec{b} & 0 \\ \vec{a}\cdot\vec{b} & |\vec{b}|^2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{4}|\vec{a}|^2|\vec{b}|^2$.

6. Area = $\int_{\sqrt{3}}^{2} x-\frac{3}{x}+\int_{2}^{3}\left(4-x-\frac{3}{x}\right)dx=\frac{4-\ln 27}{2}$.



7. If l is the length of chord, r the distance of the mid point of chord from the centre and a is radius of given circle.



r = a cos θ

l = 2a sin θ

$\frac{2}{3}$ 2a < 2a sin θ < $\frac{5}{6}$ 2a

$\frac{\sqrt{11}}{6}a < r < \frac{\sqrt{5}}{3}a$

∴ The given condition is satisfied if the mid point of the chord lies within the region between the concentric circles of radius $\frac{\sqrt{11}}{6}$ and $\frac{\sqrt{5}}{3}$.

$P = \frac{5}{9}-\frac{11}{36}=\frac{1}{4}$.

$\left(\frac{\text{Area of circular angular}}{\text{Area of given circle}}\right)$

8. Applying $R_1 \to R_1 - R_3$ and $2q = p + r$

$$\begin{vmatrix} 2^{n+1} - 2^n + p & 2^{n+2} - 2^{n+1} + q & p + r \\ 2^n + p & 2^{n+1} + q & p + r \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix} = \begin{vmatrix} 2^n + p & 2^{n+1} + q & p + r \\ 2^n + p & 2^{n+1} + q & p + r \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix} = 0 .$$

9. Point of intersection is (5, 5, 5).

10. Equation of plane containing both the lines

$$\begin{vmatrix} x-3 & y-2 & z-1 \\ 2 & 3 & 4 \\ 3 & 2 & 3 \end{vmatrix} = 0 \Rightarrow x + 6y - 5z = 10.$$

11. $({}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots.)^{n-1} = 2^{(n-1)^2}$

12. $$\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = -4$$

13. Required probability $= \frac{8}{20}.\frac{12}{19} + \frac{12}{20}.\frac{11}{19} = \frac{3}{5}$

14. Required probability $= \frac{2}{5} \cdot \frac{{}^2C_1 \cdot {}^{18}C_{10}}{{}^{20}C_{11}} \cdot \frac{1}{9} + \frac{3}{5} \cdot \frac{{}^3C_2 \cdot {}^{17}C_9}{{}^{20}C_{11}} \cdot \frac{1}{9} = \frac{99}{1900}$

15. A, B, C, D are independent events

$\therefore P(A \cap B) = \frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$ and $P(A/B) = P(A) = \frac{1}{2}$

Also $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{5} - \frac{1}{10} = \frac{3}{5}$

Now $P(A/A \cup B) = \frac{P\{A \cap (A \cup B)\}}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{1/2}{3/5} = \frac{5}{6}$.

16. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{2}{3} - P(A \cap B) = \frac{7}{6} - P(A \cap B)$

$0 \le P(A \cap B) \le \frac{1}{2}$.

17. Clearly $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{9}{2}$

$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$

$\therefore \ \vec{a} + \vec{b} + \vec{c} = 0 |\vec{a} + \vec{b} + \vec{c}|^2 \ge 0$.

19. Let OP = r

$\Rightarrow 1 = r \sin\alpha \cos\beta$ …(i)

$2 = r \sin\alpha \sin\beta$ …(ii)

$3 = r \cos\alpha$ …(iii)

Squaring and adding, we get

$r^2 = 14$

$\Rightarrow r = \pm\sqrt{14}$ …(iv)

Using (i), (ii), (iii) and (iv), we get the required result.

20. Since $\vec{a}$, $\vec{b}$ and $\vec{a}\times\vec{b}$ are non–coplanar.

So $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a}\times\vec{b})$

$\therefore\ \vec{r}\times\vec{b} = \vec{a} \Rightarrow x\vec{a}\times\vec{b} + z\left\{(\vec{a}\cdot\vec{b})\vec{b} - (\vec{b}\cdot\vec{b})\vec{a}\right\} = \vec{a}$

$\Rightarrow -\left(1 + z|\vec{b}|^2\right)\vec{a} + x\vec{a}\times\vec{b} = 0 \qquad \left[\text{since } \vec{a}\cdot\vec{b} = 0\right]$

$\therefore x = 0$ and $z = -\dfrac{1}{|\vec{b}|^2}$

Thus $\vec{r} = y\vec{b} - \dfrac{\vec{a}\times\vec{b}}{|\vec{b}|^2}$, where y is the parameter.